Trees

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Tree

A ‘free tree’ or simply a ‘tree’ is a **connected, acyclic, and undirected** graph.

Connected = There exists at least one simple path between every pair of vertices.

Ref: Section ‘B.5.1 Free trees’, CLRS 3rd edition.

Rooted Tree

A ‘rooted tree’ is a free tree in which one of the vertices is distinguished from the others. We call the distinguished vertex the ‘root’ of the tree.

Ref: Section ‘B.5.2 Rooted and ordered trees’, CLRS 3rd edition.

Binary Tree: A Recursive Definition

A ‘binary tree’ is a ***data structure*** defined on a finite set of nodes that

either contains no nodes (called an empty or null graph),

or

is composed of three disjoint sets of nodes—a root node, a **binary tree** called its left subtree, and another **binary tree** called its right subtree.

Ref: Section ‘B.5.3 Binary and positional trees’, CLRS 3rd edition.

A ***data structure*** is a way to store and organise data in order to facilitate access (read) and modifications (write).

Ref: Section ‘1.1 Algorithms’, CLRS 3rd edition.

Tree Traversals

There are three types of tree traversals or tree walks.

They are:

1. Inorder tree traversal,
2. Preorder tree traversal, and
3. Postorder tree traversal.

Ref: Section ‘12.1 What is a binary search tree?’, CLRS 3rd edition.

Inorder Tree Traversal: Pseudocode

Input: A binary tree T
Output: The in-order sequence of the keys

```
## Define a function
Inorder(vertex x)
        if x != NULL
                inorder(x.left)
                print x.key
                inorder(x.right)

// Call the function
inorder(T.root)
```

Ref: Section ‘12.1 What is a binary search tree?’, CLRS 3rd edition

Preorder Tree Traversal: Pseudocode

Input: A binary tree T
Output: The pre-order sequence of the keys

```
## Define a function
preorder(vertex x)
        if x != NULL
                print x.key
                preorder(x.left)
                preorder(x.right)

// Call the function
preorder(T.root)
```

Ref: Section ‘12.1 What is a binary search tree?’, CLRS 3rd edition

Postorder Tree Traversal: Pseudocode

Input: A binary tree T
Output: The post-order sequence of the keys

```
## Define a function
postorder(vertex x)
        if x != NULL
                postorder(x.left)
                postorder(x.right)
                print x.key

// Call the function
postorder(T.root)
```

Ref: Section '12.1 What is a binary search tree?', CLRS 3rd edition

History of the Tree Data Structure

"Pointer-based data structures [such as trees] also seem to be a folk invention. According to Knuth, pointers were apparently used in early computers with drum memories. The **A-1 language** developed by G. M. Hopper in **1951** represented algebraic formulas as binary trees".

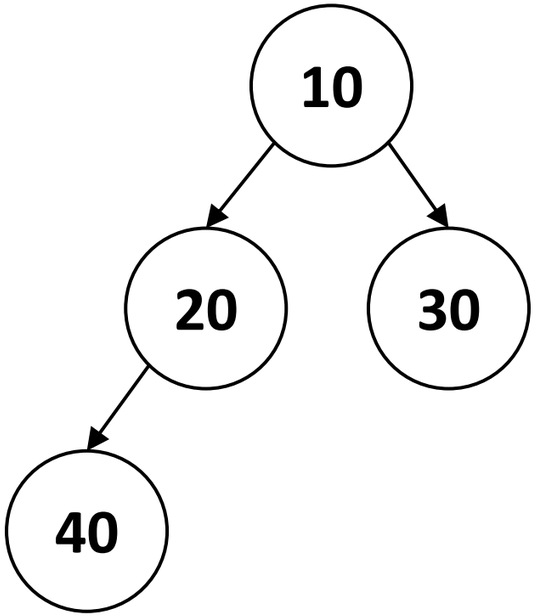
Grace Hopper (1906-1992) was a US Naval Officer, mathematician, and computer scientist. Photo courtesy: https://www.britannica.com/biography/Grace-Hopper

Ref: Chapter notes, Chapter 10 `Elementary Data Structures', CLRS 3rd edition

Practice Problem

Please write a C program to create a tree whose inorder sequence is 40 20 10 30 and the root key is 10.

Hint:



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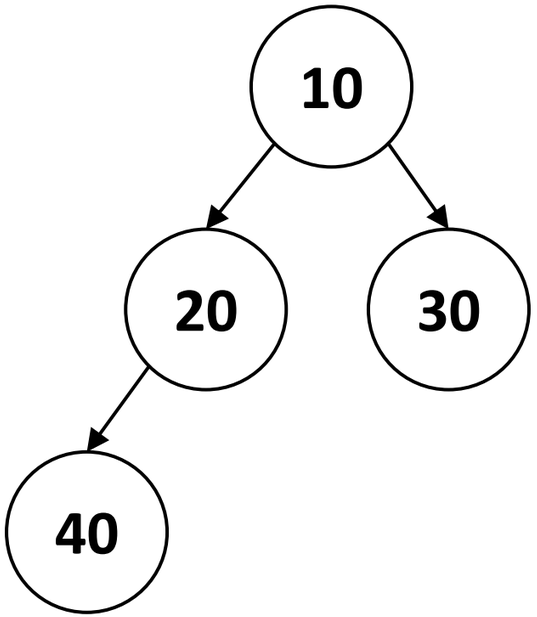

Solution. Please see ‘treeTraversal.c’ in the ‘Codes’ folder.

Thank You